

Technical Requirement Document (TRD)

CampusFind Pro - System Design and Database Architecture

1. Architecture & Platform Design

The application is built on a lightweight MVC stack utilizing native PHP 8+, MongoDB Driver extensions, and Bootstrap 5. Core operations are mapped using NoSQL document modeling to store users, reports, claims, and activity logs. The application runs locally on Apache (XAMPP) and deploys to Render Web Services behind reverse proxies.

2. MongoDB Atlas Schema Definitions

- **users:** stores email, name, avatar, verification_code (for password reset), and role ('student' or 'admin').
- **lost_items / found_items:** stores title, description, category_id, user_id (BSON ObjectId references), location, lost_date/found_date, reward details, and image filenames.
- **claims:** stores claimer_id, item_id, item_type ('lost' or 'found'), status ('pending', 'approved', or 'rejected'), and admin_remarks.
- **categories:** stores preseeded labels like Electronics, Documents, Keys, and Books.

3. Auto-Casting Database Translation Layer

To bridge the mismatch between PHP session string IDs and MongoDB BSON ObjectIds, the custom database class (config/database.php) runs a recursive query parser (castFilterIds) that automatically translates 24-character hexadecimal string keys (like user_id, category_id, _id) into BSON ObjectIds. This eliminates manual conversion boilerplate across query pages.